

Block 2 ex 2

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- 1. Compute the sample proportion of cows that survive and derive the corresponding log-odds ratio value. Compare this value with the estimate of the intercept obtained from a GLM with binomial distribution in which no predictors are used. Use the level survived as the level whose probability of occurrence is of interest.
- 2. Estimate a GLM for the variable outcome using ast as a predictor. Plot/visualize the relationship between the variables, also showing the estimated probability that a cow survives as a function of its ast level.
- 3. Now estimate a model in which the only predictor used is urea. Compare the goodness of fit of this model with that of the model estimated in point 2.
- 4. Using the model identified as best in point 3, estimate the probability that the cows with characteristics given in the dataset nd survive, and provide a 92% confidence interval for the estimated probability.

For unknown reasons, dairy cows sometimes become recumbent, meaning they lie down and are unable to rise. This condition is known as **Downer Cow Syndrome**, and these cows often suffer from a severe illness that can lead to death. The dataset '**downer.csv**' contains data from a study on blood samples collected from over 400 downer cows examined at the **Ruakura New Zealand Animal Health Laboratory** during the period **1983–1984**.

Various blood tests were performed, and for many of the animals, the outcome (survived, died, or euthanized) was recorded. The goal is to determine whether **survival can be predicted from blood measurements**. The data were collected from veterinary records, and not all variables were recorded for every cow.

The dataset (taken from *Applied Linear Regression* by S. Weisberg) contains the following variables:

- **calving**: a factor with levels *before* and *after*
- **daysrec**: Days recumbent
- **ck**: Serum creatine phosphokinase (U/l at 30°C)
- **ast**: Serum aspartate aminotransferase (U/l at 30°C)
- **urea**: Serum urea (mmol/l)
- **pcv**: Packed Cell Volume (Hematocrit)
- **inflat**: Inflammation (0 = no, 1 = yes)
- **myopathy**: Muscle disorder, a factor with levels *present* and *absent*
- **outcome**: a factor with levels *died* and *survived*

```
downer <- read.table("../Datasets/Downer.csv", sep=";", header = TRUE)
head(downer)
```

##	calving	daysrec	ck	ast	urea	pcv	inflat	myopathy	outcome
## 1	after	1	3000	590	10.9	41	no	<NA>	survived
## 2	after	4	12100	1240	23.8	NA	<NA>	present	survived
## 3	after	0	600	207	8.1	31	yes	<NA>	died
## 4	after	1	590	113	14.6	43	<NA>	<NA>	died
## 5	after	2	1800	243	12.3	48	<NA>	absent	died
## 6	before	6	380	150	NA	28	yes	absent	died

```
table(downer$outcome, useNA = "ifany")
```

```
##
##      died survived
##      269      166
```

1. Compute the sample proportion of cows that survive and derive the corresponding log-odds ratio value. Compare this value with the estimate of the intercept obtained from a GLM with binomial distribution

in which no predictors are used. Use the level survived as the level whose probability of occurrence is of interest.

```
p_hat <- mean(downer$outcome == "survived")
p_hat
```

```
## [1] 0.3816092
```

```
logit_p <- log(p_hat / (1 - p_hat))
logit_p
```

```
## [1] -0.4827236
```

```
downer$out <- I(downer$outcome == "survived")

fit0 <- glm(out ~ 1, family = binomial, data = downer)
summary(fit0)
```

```
##
## Call:
## glm(formula = out ~ 1, family = binomial, data = downer)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.4827      0.0987  -4.891   1e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 578.42  on 434  degrees of freedom
## Residual deviance: 578.42  on 434  degrees of freedom
## AIC: 580.42
##
## Number of Fisher Scoring iterations: 4
```

```
coef(fit0)[1]
```

```
## (Intercept)
##  -0.4827236
```

2. Estimate a GLM for the variable outcome using ast as a predictor. Plot/visualize the relationship between the variables, also showing the estimated probability that a cow survives as a function of its ast level.

```
fit1 <- glm(out ~ ast, family = binomial, data = downer)
summary(fit1)
```

```
##
## Call:
## glm(formula = out ~ ast, family = binomial, data = downer)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.5222213  0.1695120   3.081  0.00206 **
## ast         -0.0032394  0.0005198  -6.232  4.6e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 569.75  on 428  degrees of freedom
## Residual deviance: 499.84  on 427  degrees of freedom
##    (6 osservazioni eliminate a causa di valori mancanti)
## AIC: 503.84
##
## Number of Fisher Scoring iterations: 5
```

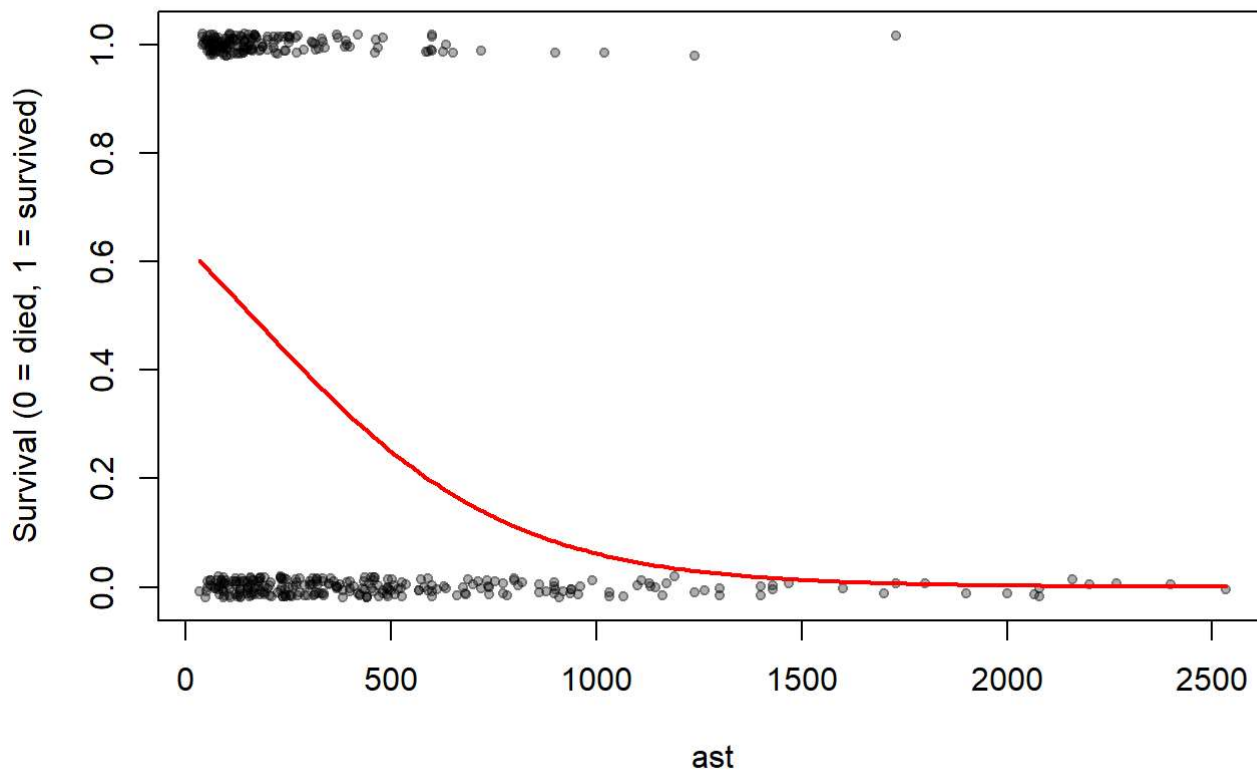
```
ast_seq <- seq(min(downer$ast, na.rm = TRUE),
               max(downer$ast, na.rm = TRUE),
               length.out = 200)

newdat <- data.frame(ast = ast_seq)

newdat$phat <- predict(fit1, newdata = newdat, type = "response")

plot(downer$ast, jitter(as.numeric(downer$out), amount=0.02),
     xlab = "ast",
     ylab = "Survival (0 = died, 1 = survived)",
     pch = 20, col = rgb(0, 0, 0, 0.3))

lines(newdat$ast, newdat$phat,
      col = "red", lwd = 2)
```



3. Now estimate a model in which the only predictor used is urea. Compare the goodness of fit of this model with that of the model estimated in point 2.

```
fit2 <- glm(out ~ urea, family = binomial, data = downer)
summary(fit2)
```

```
##
## Call:
## glm(formula = out ~ urea, family = binomial, data = downer)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.60434    0.28417   2.127  0.0334 *
## urea        -0.12116    0.03102  -3.906 9.39e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 354.17  on 265  degrees of freedom
## Residual deviance: 331.11  on 264  degrees of freedom
## (169 osservazioni eliminate a causa di valori mancanti)
## AIC: 335.11
##
## Number of Fisher Scoring iterations: 5
```

Both models show a significant negative effect of the predictor (higher values -> lower survival probability), but the model with urea has a clearly smaller residual deviance and AIC, so—among the cows with non-missing urea—it provides a **better** fit to the data than the model with ast.

4. Using the model identified as best in point 3, estimate the probability that the cows with characteristics given in the dataset `nd` survive, and provide a 92% confidence interval for the estimated probability.

```
nd <- data.frame(ast = c(500, 1400, 2000),  
  urea = c(5, 20, 30))  
  
pred <- predict(fit2, newdata=nd, type = "link", se.fit = TRUE)  
  
fit <- pred$fit  
se.fit <- pred$se.fit  
  
cbind(binomial()$linkinv(fit + qnorm(.04) * se.fit),  
  binomial()$linkinv(fit + qnorm(.96) * se.fit))
```

```
##           [,1]      [,2]  
## 1 0.42873915 0.5705521  
## 2 0.07563871 0.2433379  
## 3 0.01420190 0.1393544
```